

10. Prove Theorem 2:

THEOREM 2

Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1r - c_2 = 0$ has only one root r_0 . A sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if $a_n = \alpha_1r_0^n + \alpha_2nr_0^n$, for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants.

42. Show that if $a_n = a_{n-1} + a_{n-2}$, $a_0 = s$ and $a_1 = t$, where s and t are constants, then $a_n = sf_{n-1} + tf_n$ for all positive integers n .
46. Suppose that there are two goats on an island initially. The number of goats on the island doubles every year by natural reproduction, and some goats are either added or removed each year.
- Construct a recurrence relation for the number of goats on the island at the start of the n th year, assuming that during each year an extra 100 goats are put on the island.
 - Solve the recurrence relation from part (a) to find the number of goats on the island at the start of the n th year.
 - Construct a recurrence relation for the number of goats on the island at the start of the n th year, assuming that n goats are removed during the n th year for each $n \geq 3$.
 - Solve the recurrence relation in part (c) for the number of goats on the island at the start of the n th year.